TOC-2025: Quiz-2 (DFA Construction)

Full Marks: 15

(1) Let $D = \{w \in \{0,1\}^* \mid \text{every odd position of } w \text{ is a } 1\}$.

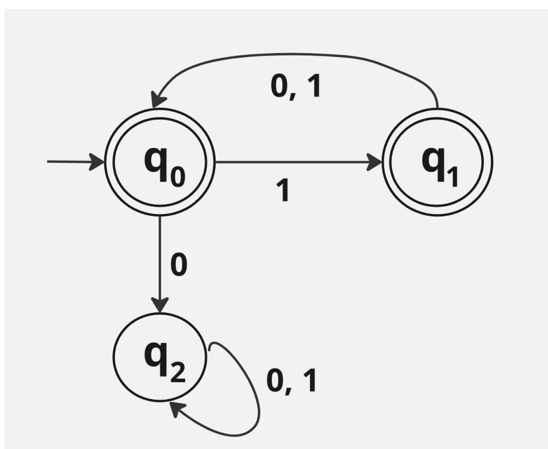
(a) Design a Deterministic Finite Automata (DFA) M for D.

(b) Formally prove that $L(M) = D$.

(**5 Marks + 10 Marks**)

Solution: The answer has two parts.

(a) We design a DFA $M = (Q, \Sigma, \delta, q_0, F)$ is defined by the following transition diagram.



In this diagram $F = \{q_0, q_1\}$.

(b) Proof of $L(M) = D$.

In order to give a proof of both directions of the subset containment, we prove the following statement.

<u>Statement:</u> For every $w \in D$, the followings are satisfied.

(i) if $|w|$ is even, then $\widehat{\delta}(q_0, w) = q_0$.

(ii) if $|w|$ is odd, then $\widehat{\delta}(q_0, w) = q_1$.

Basis Step: $|w| = 0$ and $|w| = 1$ are two base cases.

If $|w| = 0$, then $w = \varepsilon$.

Then, $\widehat{\delta}(q_0, w) = \widehat{\delta}(q_0, \varepsilon) = q_0$.

If $|w| = 1$, then $w = 1$ as that is the only possible string of length exactly one in D.

Then, $\widehat{\delta}(q_0, 1) = q_1$.

Induction Hypothesis: For every string $y \in D$ of length at most k.

(i) if $|y|$ is even, then $\widehat{\delta}(q_0, y) = q_0$.

(ii) if $|y|$ is odd, then $\widehat{\delta}(q_0, y) = q_1$.

Induction Step: Consider a string $w \in D$ such that $|w| = k + 1$.

Suppose that $w = xb$ such that $b \in \{0, 1\}$.

A crucial observation is that since $w \in D$, it follows that $x \in D$.

Then, $\widehat{\delta}(q_0, w) = \widehat{\delta}(q_0, xb) = \delta(\widehat{\delta}(q_0, x))$.

If $|w|$ is even, then $|x|$ is odd.

As $x \in D$, due to induction hypothesis $\widehat{\delta}(q_0, x) = q_1$.

Since $\delta(q_1, b) = q_0$, therefore, $\widehat{\delta}(q_0, w) = q_0$. This completes the proof of (i).

Similarly, if $|w|$ is odd, then $|x|$ is even.

Similarly, as $x \in D$, due to induction hypothesis, $\widehat{\delta}(q_0, x) = q_0$.

Hence, $\widehat{\delta}(q_0, xb) = \delta(q_0, b)$.

But what is crucial is that the last symbol b appears in an odd position. Hence, $b = 1$.

Then, $\delta(q_0, b) = q_1$. Hence, (ii) is true.

Now, using the above statement, we prove both the directions as follows.

First, we prove the <u>backward direction</u> ($\Leftarrow$) which says that "if $w \in D$, then $w \in L(M)$".

Since (i) and (ii) are true, then for every $w \in D$, $\widehat{\delta}(q_0, w) \in \{q_0, q_1\}$.

Hence, $w \in L(M)$.

Now, we give the <u>forward direction</u> ($\Rightarrow$) of the proof.

Let $w \in L(M)$.

Suppose for the sake of contradiction that there is an odd position of w that has symbol 0.

Then, consider $(2p + 1)$, the smallest odd numbered index such that $(2p + 1)$-th character is 0.

Then, $w = x0y$ such that $|x|$ is even, every odd position of x has 1, and $y \in \Sigma^*$.

Then, $\widehat{\delta}(q_0, x) = q_0$.

Subsequently, $\delta(q_0, 1) = q_2$ and for any $y \in \Sigma^*$, it holds that $\widehat{\delta}(q_2, y) = q_2$.

Then, $\widehat{\delta}(q_0, x0y) \notin F$ implying that $x0y (= w) \notin L(M)$.

This completes the proof of the backward direction.